AQ7.3: Activity Questions 3 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

If $||a|| = 3$, $||b|| = 2\sqrt{2}$ and angle between a and b is 45°, then the value of $a.b$ is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point

1 point
Given that $\| a \| = 3$ and $\| b \| = 5$ and $a \cdot b = 7.5$, for two vectors a and b in R^n. Find the angle (in degrees) between the two vectors a, b. Note: $\cos 60^\circ = 1/2, \cos 30^\circ = \sqrt{3}/2$.

- 30°
- 60°
- 120°
- 150°

No, the answer is incorrect.
Score: 0
Accepted Answers:
60°
1 point

Consider two vectors $a = (3, 4, -1), b = (2, -1, 2)$ in R^3. Choose the correct options.

- [ ] Length/Norm of a is $\sqrt{26}$ and that of b is 3.
- [ ] Length/Norm of a is 5 and that of b is 3.
- [ ] Length/Norm of a is $\sqrt{24}$ and that of b is $\sqrt{7}$.
- [ ] Angle between the two vectors is 90 degrees.

☐
Angle between the two vectors is approximately 00 degrees.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Length/Norm of aa is $\sqrt{26}$ 26
√
and that of bb is 33.
Angle between the two vectors is 9090 degrees.

1 point

If the dot product of two vectors is zero, then choose the correct option.

- ○ The two vectors are perpendicular to each other.
- ○ The two vectors are parallel to each other.
- ○ The two vectors are exactly the same.
- ○ The length of the two vectors must be the same.

No, the answer is incorrect.
Score: 0
Accepted Answers:
The two vectors are perpendicular to each other.

Level 2

If $a = (6, -1, 3), b = (4, c, 2)$a = (6, −1, 3), b = (4, c, 2) where $a, b \in R^3$ a, b ∈ R3 and aa and bb are perpendicular to each other, then find the value of cc?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 30

1 point
1 point

Consider two vectors $a = (1, 2), b = (2, 2)$a = (1, 2), b = (2, 2) and θθ is the angle between them. The sum of the two vectors is given by $c = a + b$c = a + b. Choose the correct options.

☐
Length/Norm of cc is 55.

☐
Length/Norm of cc is 2525.

☐
Length/Norm of cc is 44.

☐
$\cos\theta = \frac{3}{\sqrt{10}}$cos θ = 10
√
3.

☐
$\cos\theta = \frac{3}{\sqrt{5}}$cos θ = 5
√
3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

Length/Norm of c is 5.

$\cos\theta = \frac{3}{\sqrt{10}}$

3.

1 point

Consider 3 vectors a, b, c in R^3 and a scalar λ in R. Choose the set of correct options.

Note: $(.)$ represents the dot product.

- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot b$
- [ ] $\lambda(a \cdot b) = (\lambda b) \cdot a$
- [ ] $\lambda(a \cdot b) = (\lambda a) \cdot (\lambda b)$
- [ ] $(a + c) \cdot b = a \cdot b + c \cdot b$
- [ ] $(a + c) \cdot b = a \cdot c + c \cdot b$
- [ ] $a \cdot a = 0, b \cdot b = 0$ if and only if a, b are null vectors.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\lambda(a \cdot b) = (\lambda a) \cdot b$

$\lambda(a \cdot b) = (\lambda b) \cdot a$

$(a + c) \cdot b = a \cdot b + c \cdot b$

$a \cdot a = 0, b \cdot b = 0$ if and only if a, b are null vectors.

1 point

Consider two vectors $a = (1, 3, 5, 7, 9)$ and $b = (2, 4, 6, 8, 10)$. Choose the set of correct options.

- [ ] The length of b is more than a.
- [ ] The length of a is more than b.
- [ ] The length of $(a - b)$ is $\sqrt{5}$.
- [ ] The length of $(a - b)$ is $\sqrt{50}$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The length of b is more than a.

The length of $(a - b)$ is $\sqrt{5}$.

Check Answers

Your score is: 0/8